Surrogate models for FE^2: Classic mesoscale modeling, pre-trained data-driven models, physics-informed subspace projection and probabilistic active learning

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Vision: enable multiscale analysis with data-driven tools

Calibration

Hyper-reduction

Offline

Pre-trained NN

Physics

?

Data

FE^2

Adaptive hyper-reduction

Online

Adaptive GP

Offline

Physics Data

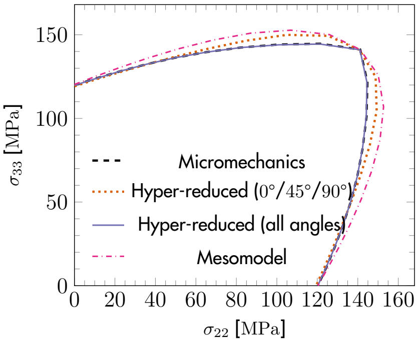
Online

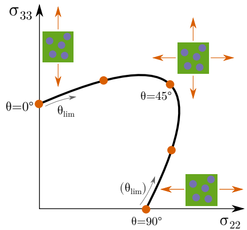
Mesomodel, hyper-reduction, neural networks

Predicting the complete biaxial transverse tension yield envelope:

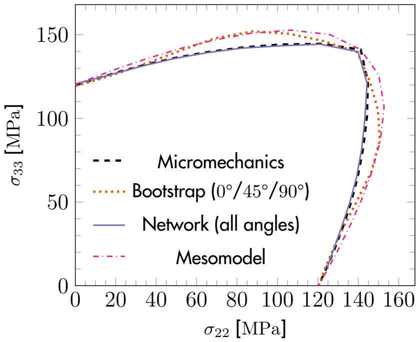
- With enough training, both models outperform a state-of-the-art plasticity mesomodel

Hyper-reduced





Neural network

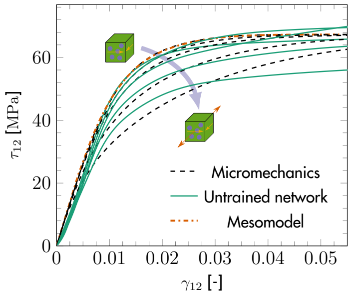


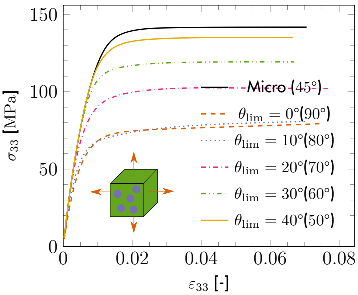
[Rocha et al. (2020), *Eur J Mech A/Sol*, **82:**103995**]**

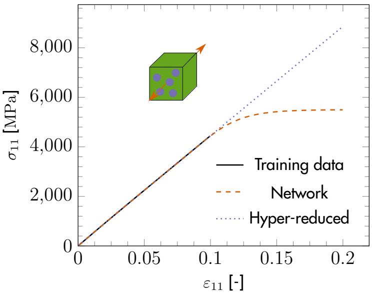
The extrapolation problem

The consequence of discarding physics:

- Senseless results out of training strain range, especially for purely data-driven models
- Training for every possible load case is intractable. Adaptivity is key!





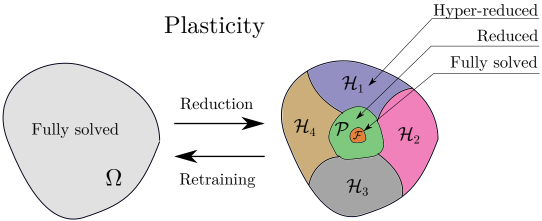


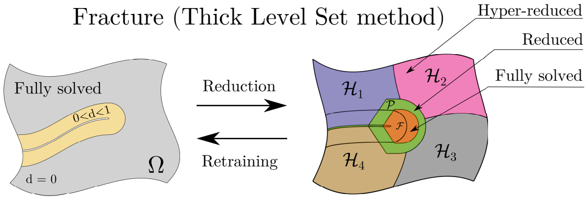
[**Rocha et al. (2020)**, *Eur J Mech A/Sol*, **82**:103995]

Active learning with hyper-reduction

Adaptive hyper-reduction scheme without offline training:

- First step fully solved ⇒ POD reduction ⇒ Hyper-reduction
- Adaptivity: Deviation from equilibrium in original space ⇒ retraining





[**Rocha, vd Meer, Sluys (2020),** *Comput Methods Appl Mech Eng*, **358:**112650]

[**Rocha, vd Meer, Mororó, Sluys (2020),** *Int J Numer Meth Eng*, **121:**2147-2173]